Abdul Rafi

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Profile

Computer Science and Engineering undergraduate at the University of Moratuwa with a strong interest in software engineering, distributed systems, full-stack development, and intelligent systems. I enjoy designing and building reliable software from requirements and system architecture through implementation, integration, and deployment. My experience includes full-stack web and mobile applications, backend APIs, databases, event-driven systems. I am particularly interested in developing scalable, maintainable software systems and combining software engineering with AI and computer vision to solve real-world problems.

Education

University of Moratuwa, Sri Lanka March 2024 – Present
B.Sc. Engineering (Hons.) in Computer Science & Engineering
Specialization in Integrated Computer Engineering CGPA: 3.15 / 4.00

St. Thomas' College, Matara January 2009 – January 2023
G.C.E. Advanced Level (Physical Science Stream)

- Achieved 3 A grades (Combined Mathematics, Physics, Chemistry)
- All-Island Rank: 206 | Z-Score: 2.5139

Engineering Projects

School Management System GitHub
Real Client Project | *React* | *FastAPI* | *Supabase* | *PostgreSQL* | *Mobile + Push Notifications*
- Built a full-stack attendance and student management platform (React + TypeScript, React Native, FastAPI, Supabase/PostgreSQL) for a school of approximately 700 students, used in production by real teachers and staff.
- Implemented role-based access, scheduled push notifications, and automated Excel/PDF reporting; owned the project end-to-end from requirements to deployment.

Kandypack Distribution System GitHub
Semester 03 Group Project | *React* | *Node.js/Express* | *MySQL*
- Built a full-stack logistics management system for a hybrid rail-and-road supply chain, replacing manual Excel-based tracking with a relational database, REST API, and React dashboard.
- Designed a MySQL schema with triggers and stored procedures to enforce business rules: order lead-time validation, train capacity limits, staff working-hour caps, and schedule conflict prevention.
- Implemented JWT-based role-based access (Admin/Driver/Assistant), automated order-to-train-trip allocation, and reporting on sales, driver hours, and truck usage.

OnTime Distributed Transit System GitHub • Messaging Epic
Semester 04 Group Project | *Distributed Systems* | *Kafka* | *MQTT* | *FastAPI* | *Redis* | *Kubernetes*
- Developed the messaging backbone for a distributed transit platform using HiveMQ MQTT, Apache Kafka, WebSockets, and Redis for real-time telemetry and inter-service communication.
- Configured Kafka in KRaft mode with partitioned and replicated topics, and integrated an MQTT-to-Kafka ingestion pipeline with a FastAPI WebSocket service for live data delivery.
- Contributed to Docker/Kubernetes-based deployment, messaging infrastructure integration, smoke testing, and end-to-end verification of the real-time telemetry pipeline.

Easy Pharma – Prescription OCR Pipeline GitHub
Healthcare AI | *OCR* | *React PWA* | *FastAPI* | *OpenCV* | *Tesseract* | *AWS Textract*
- Built a prescription-scanning web application that converts prescription images into structured, reviewable medicine lists, using a React + TypeScript + Vite PWA with camera capture, mobile-responsive design, and offline app-shell support.
- Developed a FastAPI backend with an OpenCV preprocessing pipeline (orientation correction, resizing, optional deskewing, adaptive thresholding, denoising) feeding local Tesseract OCR, with an optional AWS Textract fallback for low-confidence scans.
- Implemented a parser for dosage values (mg, mcg, g, ml) and RapidFuzz-based medicine matching against a dataset to correct OCR spelling errors; added evaluation utilities for CER/WER, structured recognition, and end-to-end latency.

Technical Skills

Programming Languages – Python, Java, JavaScript, TypeScript, C++, C

Web Development – React, React Native, Node.js/Express, FastAPI, REST APIs, JWT Authentication

Databases – MySQL, PostgreSQL, MongoDB, Supabase

Distributed Systems & Messaging – Apache Kafka (KRaft), MQTT, WebSockets, Redis

Cloud & DevOps – Docker, Kubernetes, Terraform, Linux (Ubuntu)

AI & Computer Vision (Working Knowledge) – OpenCV, LangChain, LLMs, RAG, YOLO

Engineering Tools – Git, CMake

Leadership & Service

- Programme Committee Member – Hit the Ground Tournament, University of Moratuwa
- Logistics Committee Member – SLIoT Competition, University of Moratuwa
- Logistics & Marketing Team Member – Career Fair 26
- Religious Coordinator – Majlis-Ul-Islam, University of Moratuwa

Achievements

- Runner-up – IEEE INSL Idea Stage 2025 Competition
- Finalist – YARL Geek Challenge Season 14

References

Prof. Chandana D. Gamage
Professor
University of Moratuwa, Sri Lanka
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